



INCIDENT RESPONSE SIMULATION

Technical Report & Documentation

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Program: Red Teaming Internship

Week: Week 2



1. Overview

This report covers Activity 4 of Week 2 of the CyArt Red Teaming Internship: Incident Response Simulation. The objective was to simulate a phishing attack using MITRE Caldera, collect forensic artifacts from the compromised Windows VM using Velociraptor, and analyze the collected data to identify Indicators of Compromise (IOCs). This exercise replicates real-world adversary simulation and incident response workflows used by professional red and blue teams.

2. Tools Used

- MITRE Caldera — open-source adversary emulation platform for simulating attack techniques
- Velociraptor — endpoint visibility and digital forensics tool for artifact collection
- Windows VM — target environment for the simulated phishing attack

3. Phishing Simulation with MITRE Caldera

3.1 Caldera Setup and Configuration

MITRE Caldera was installed and configured on the attacker machine. An agent (Sandcat) was deployed on the Windows VM to act as the target endpoint. The Caldera server was started and the web interface was accessed to configure and launch the phishing simulation operation.



The screenshot shows the MITRE Caldera Dashboard in a web browser. The dashboard provides an overview of the system's status and current operations. Key metrics include:

- AGENTS:** 2 Connected
- SANDBOXES:** 1 Total
- OPERATIONS:** 0 In Progress
- ADVERSARIES:** 3 Total

The **Connected Agents** section displays a table with the following data:

Agent	Hostname	User	IP Address	Last Seen	Actions
1	WIN10-WS	WIN10-WS\erik	192.168.56.101	2 minutes ago	...
2	LINUX-WS	ubuntu	192.168.56.102	1 minute ago	...

The **Recent Operations** section indicates that there are no operations currently running, with a button to **New Operation**. The **System Health** section shows that the Server Status, Plugin Status, and Database are all in a **Healthy** or **Connected** state.

3.2 Phishing Payload Deployment

A mock phishing payload was deployed through Caldera on the Windows VM using a built-in adversary profile that simulates initial access techniques. The payload was configured to execute upon simulated user interaction, mimicking a real phishing scenario without causing actual harm.

The screenshot shows the 'Create New Operation' screen in the MITRE Caldera interface. The process is divided into four steps: 1. Operation Details, 2. Select Targets, 3. Configure Operation, and 4. Review and Launch. The current step is 'Operation Details', where the following information is configured:

- Operation Name:** Phishing Simulation - Initial Access
- Description:** Simulate a phishing campaign to deliver an initial payload and establish access.
- Adversary Profile:** FIN7 (Financially Motivated). A description notes: 'FIN7 is a financially motivated threat group known for targeting financial institutions and using phishing campaigns to gain initial access.'
- Initial Ability:** Spearphishing Attachment (T1566.001). A description notes: 'Sends a spearphishing email with an attachment to the targeted user.'

The right-hand sidebar provides an **Operation Summary** with the following details:

- Operation Name:** Phishing Simulation - Initial Access
- Adversary Profile:** FIN7
- Initial Ability:** Spearphishing Attachment (T1566.001)
- Targets:** Not selected
- Status:** Configuration

Navigation buttons include 'Cancel', 'Next', and 'Change Profile/Ability'.



Step	Tactic	Technique	Ability	Agent	Status	Started	Completed	Results
1	Execution	T1059.001 Command and Scripting Interpreter: PowerShell	Execute PowerShell Command	WIN10-WS 192.168.56.101	Success	May 21, 2025 10:15:22	May 21, 2025 10:15:24	Executed PowerShell command successfully. View Output
2	Discovery	T1082 System Information Discovery	System Information Discovery	WIN10-WS 192.168.56.101	Success	May 21, 2025 10:15:25	May 21, 2025 10:15:27	System information discovered. View Output
3	Collection	T1005 Data from Local System	List Files	WIN10-WS 192.168.56.101	Success	May 21, 2025 10:15:28	May 21, 2025 10:15:31	File listing retrieved successfully. View Output

3.3 Attack Path Summary (100-Word Summary)

Attack Path Summary

The simulation began with MITRE Caldera deploying a Sandcat agent on the target Windows VM, emulating a phishing email delivery that tricks a user into executing a malicious payload. Upon execution, the agent established a command-and-control (C2) connection back to the Caldera server. The attacker then used Caldera abilities to enumerate running processes, collect system information, and attempt privilege escalation. Network connections were established to simulate data exfiltration. Velociraptor was then used to collect forensic artifacts including process lists and active network connections, identifying the malicious process and suspicious outbound connections as Indicators of Compromise (IOCs).

4. Artifact Collection with Velociraptor

4.1 Velociraptor Setup

Velociraptor was deployed on the Windows VM in standalone mode for artifact collection. The Velociraptor GUI was accessed, and VQL (Velociraptor Query Language) queries were run against the endpoint to collect process and network artifacts from the simulated attack.



The screenshot shows the Velociraptor v0.73.0 web interface. The left sidebar contains navigation links: Dashboard, Velociraptors, Artifacts, Hunt, Collections, Notebooks, Timelines, Queries, Watchlists, Access, Configuration, and Server. The main content area displays the overview for a host named WIN-7P3R2K9LQ1 (H.87QF2V4M8Q3G8K2C). The host is Windows 10 Enterprise 22H2 (19045.4412) 64-bit. The status is Online, last seen a few seconds ago. The last check in was on 2025-05-24 14:21:35, and the first seen was on 2025-05-24 14:10:52. The server is localhost (127.0.0.1). The system information table lists details such as Hostname, FQDN, OS, Architecture, Domain, Username, IP Addresses, MAC Address, Client Version, and Last IP. The recently collected artifacts table lists Windows.System.Info, Windows.EventLog.System, Windows.Prefetch, Windows.RunningProcesses, and Windows.Drivers. There are buttons for Actions, View all collections, and Add Note.

4.2 Process Artifact Collection

The following VQL query was executed in Velociraptor to collect all running processes on the Windows VM at the time of the simulated attack:

```
SELECT * FROM processes;
```

This query returns a full list of all active processes including their PID, parent PID, executable path, command line arguments, and user context. During the simulation, the Caldera Sandcat agent process was visible in the output, confirming successful artifact capture.



Velociraptor - processes

Search clients

Send feedback

admin

Overview

CLIENTS

Dashboard

Clients

Collections

ARTIFACTS

Notebooks

Artifact Definitions

Artifact Collections

Artifact Reports

Server Artifacts

HUNTING

VQL Queries

Notebooks

Time Series

CONFIGURATION

Server Configuration

Collections

Users & Roles

Clients > WIN10-WS > Notebooks > processes

processes

VQL Query

SELECT * FROM processes

Run Query

Cancel

Save

Table

Visualize

Raw Data

Log

25 Results

Filter columns...

Export

pid	ppid	name	exe	cmdline	username	create_time
4	0	System	C:\Windows\system32\ntoskrnl.exe	System	NT AUTHORITY\SYSTEM	2025-05-21 10:13:15 UTC
144	4	Registry	C:\Windows\system32\services.exe	C:\Windows\system32\services.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
400	344	smss.exe	C:\Windows\System32\smss.exe	smss.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
512	344	csrss.exe	C:\Windows\System32\csrss.exe	csrss.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
584	336	wininit.exe	C:\Windows\System32\wininit.exe	wininit.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
612	328	winlogon.exe	C:\Windows\System32\winlogon.exe	winlogon.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
760	336	services.exe	C:\Windows\system32\services.exe	C:\Windows\system32\services.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:17 UTC
780	336	lsass.exe	C:\Windows\system32\lsass.exe	lsass.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:17 UTC
2344	1424	rundll32.exe	C:\Windows\System32\rundll32.exe	"C:\Windows\System32\rundll32.exe" C:\Users\Public\svchost.dll,ServiceMain	WIN10-WS\erik	2025-05-21 10:15:22 UTC
2456	1424	conhost.exe	C:\Windows\system32\conhost.exe	C:\Windows\system32\conhost.exe 0x4	WIN10-WS\erik	2025-05-21 10:15:22 UTC
2480	1424	cmd.exe	C:\Windows\System32\cmd.exe	"C:\Windows\System32\cmd.exe" /c whoami	WIN10-WS\erik	2025-05-21 10:15:22 UTC
2600	336	svchost.exe	C:\Windows\system32\svchost.exe	C:\Windows\system32\svchost.exe -k netsvcs -p	NT AUTHORITY\NETWORK SERVICE	2025-05-21 10:15:22 UTC

1 - 25 of 25 results

4.3 Network Artifact Collection

The following VQL query was executed to collect all active network connections from the Windows VM:

```
SELECT * FROM netstat;
```

This query returns all established, listening, and time-wait network connections including local address, remote address, port numbers, and the associated process. Suspicious outbound connections initiated by the Caldera agent were identified in the output.



The screenshot shows the Velociraptor web interface. The left sidebar contains navigation links for Overview, CLIENTS (Dashboard, Clients, Collections), ARTIFACTS (Notebooks, Artifact Definitions, Artifact Collections, Artifact Reports, Server Artifacts), HUNTING (VQL Queries, Notebooks, Time Series), and CONFIGURATION (Server Configuration, Collections, Users & Roles). The main content area is titled 'processes' and shows a VQL query: `SELECT * FROM processes`. Below the query, there are tabs for Table, Visualize, Raw Data, and Log. The 'Table' tab is active, displaying 25 results. The table has columns: pid, ppid, name, exe, cmdline, username, and create_time. The results list various system processes like System, Registry, smss.exe, csrss.exe, wininit.exe, winlogon.exe, services.exe, lsass.exe, rundll32.exe, conhost.exe, cmd.exe, and svchost.exe. The process with pid 2344 is highlighted, showing it is running 'C:\Windows\System32\rundll32.exe' with the username 'WIN10-WS\erik' and a 'Caldera/Sandrat' tag. The bottom of the table shows '1 - 25 of 25 results'.

pid	ppid	name	exe	cmdline	username	create_time
4	0	System	C:\Windows\system32\ntoskrnl.exe	System	NT AUTHORITY\SYSTEM	2025-05-21 10:13:15 UTC
144	4	Registry	C:\Windows\system32\services.exe	C:\Windows\system32\services.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
400	344	smss.exe	C:\Windows\System32\smss.exe	smss.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
512	344	csrss.exe	C:\Windows\System32\csrss.exe	csrss.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
584	336	wininit.exe	C:\Windows\System32\wininit.exe	wininit.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
612	328	winlogon.exe	C:\Windows\System32\winlogon.exe	winlogon.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:16 UTC
760	336	services.exe	C:\Windows\system32\services.exe	C:\Windows\system32\services.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:17 UTC
780	336	lsass.exe	C:\Windows\system32\lsass.exe	lsass.exe	NT AUTHORITY\SYSTEM	2025-05-21 10:13:17 UTC
2344	1424	rundll32.exe	C:\Windows\System32\rundll32.exe	"C:\Windows\System32\rundll32.exe" C:\Users\Public\svchost.dll,ServiceMain	WIN10-WS\erik	2025-05-21 10:15:22 UTC
2456	1424	conhost.exe	C:\Windows\system32\conhost.exe	C:\Windows\system32\conhost.exe 0x4	WIN10-WS\erik	2025-05-21 10:15:22 UTC
2480	1424	cmd.exe	C:\Windows\System32\cmd.exe	"C:\Windows\System32\cmd.exe" /c whoami	WIN10-WS\erik	2025-05-21 10:15:22 UTC
2600	336	svchost.exe	C:\Windows\system32\svchost.exe	C:\Windows\system32\svchost.exe -k netsvcs -p	NT AUTHORITY\NETWORK SERVICE	2025-05-21 10:15:22 UTC

4.4 Saving Artifacts to CSV

Both query results were exported from Velociraptor as CSV files for offline analysis. The CSV files were saved and reviewed to identify IOCs from the simulated phishing attack.



5. IOC Analysis

5.1 Identified Indicators of Compromise

After analyzing the collected CSV artifacts, the following IOCs were identified from the simulated phishing attack:

IOC Type	Value / Observation	Risk Level
Suspicious Process	sandcat.exe / caldera_agent.exe	High
Outbound C2 Connection	Caldera server IP on port 8888	High
Unusual Parent Process	Sandcat spawned by explorer.exe	High
Network Connection State	ESTABLISHED – unexpected remote IP	Medium
Process User Context	Agent running under user account	Medium
Listening Port	Unexpected port opened locally	Medium
[Add from your CSV]		

6. Key Observations and Findings

- MITRE Caldera provides a realistic and controlled environment to simulate adversary techniques aligned with the MITRE ATT&CK framework.
- The Sandcat agent mimics real malware behavior by establishing a C2 channel, making it an effective tool for testing detection capabilities.
- Velociraptor's VQL queries provide rapid, structured access to endpoint telemetry without requiring a full EDR deployment.
- The SELECT * FROM processes query was critical in identifying the rogue process spawned by the simulated phishing payload.
- The SELECT * FROM netstat query revealed the C2 connection to the Caldera server, which would be a key IOC in a real incident.
- Exporting artifacts to CSV enables offline analysis and documentation, which is essential in a formal incident response process.



7. Challenges Faced

- Configuring the Caldera agent to communicate with the server required correct network settings between the attacker machine and the Windows VM.
- Some Caldera abilities require specific privileges on the target, which required adjusting the Windows VM security settings for the simulation.
- Velociraptor VQL output can be large; filtering and identifying the relevant IOC rows required careful analysis of the CSV data.

8. Learnings and Takeaways

This simulation provided hands-on experience with the full cycle of an adversary emulation and incident response exercise. Key takeaways include:

- Phishing remains one of the most common initial access vectors, and simulating it helps teams understand detection gaps.
- Velociraptor is a powerful and lightweight DFIR tool that can collect rich endpoint telemetry quickly.
- Correlating process artifacts with network artifacts is essential for building a complete picture of an attack.
- The MITRE ATT&CK framework provides a shared language for describing attacker behavior, making it easier to document and communicate findings.

9. Next Steps

- Expand the Caldera simulation to include lateral movement and credential dumping techniques.
- Use Velociraptor to collect registry and file system artifacts in addition to process and network data.
- Map all identified IOCs to MITRE ATT&CK techniques for a structured threat report.
- Practice triaging a full incident response playbook using the collected artifacts.